Instructions, User Manuals, and Standard Operating Procedures

Why Write an Instruction? Include instructions or user manuals whenever your audience needs to know how to

- Operate a mechanism
- Collect lab specimens
- Install equipment
- Use an autoclave
- Manufacture a product
- Service equipment
- Package a product
- Troubleshoot a system
- Perform lab experiments
- Use software
- Test components
- Set up a product
- Maintain equipment
- Implement a procedure
- Clean a product
- Assemble a product
- Draw blood
- Repair a system

Criteria for Writing Instructions

Audience Recognition

As the writer, you should provide your readers with the clarity and thoroughness they require.

The key to success as a writer of instructions is the following: Don't assume anything. Spell it all out—clearly and thoroughly.

Ethical Instructions

Legalities in User Manuals.

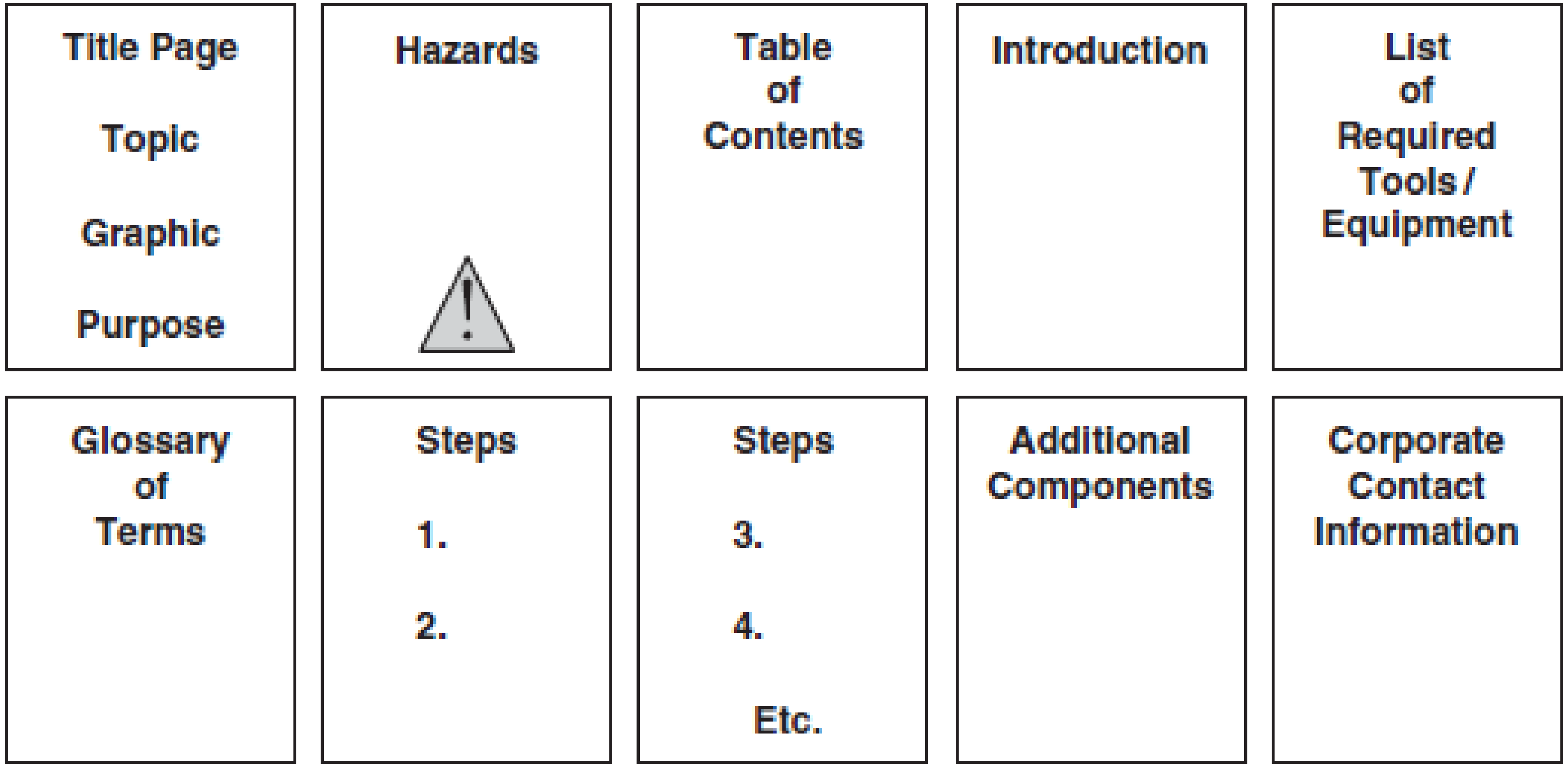
Clearly stated **hazard notations, warranties, and disclaimers** allow you to warn your audience of potential dangers and to set limits and exceptions for product guarantees.

By doing so, you adhere to your legal responsibilities to the company and to the client.

Practicalities in User Manuals.

Components of Instructions

FIGURE 2 Key Components of Instructional Manuals



Components of Instructions

Title Page-It consists of topic, purpose, graphic depiction

Safety Requirements-Correctly using hazard notations will avoid expensive liability issues and adhere to ethical communication standards.

Access-To make the hazard notations obvious, vary your typeface and type size, use white space to separate the warning or caution from surrounding text, box the warning or caution, and call attention to the hazards through graphics.

Components of Instructions

Definitions.

To avoid confusion, we suggest the following hierarchy of definitions, which clarifies the degree of hazard:

Note. Important information, necessary to perform a task effectively or to avoid loss of data or inconvenience.

Caution. The potential for damage or destruction of equipment.

Warning. The potential for serious personal injury.

Danger. The potential for death.

Components of Instructions

- **Colors.** Another way to emphasize your hazard message is through a colored window or text box around the word.
- **Usually, *Note* is printed in blue or black, *Caution* in yellow, *Warning* in orange, and *Danger* in red.**
- **Text.** To further clarify your terminology, provide the readers text to accompany your hazard alert. Your text should have the following three parts:

A one- or two-word identification alerting the reader. Words such as “High Voltage,” “Hot Equipment,” “Sharp Objects,” or “Magnetic Parts,” for example, will warn your reader of potential dangers, warnings, or cautions.

The consequences of the hazards, in three to five words. Phrases like “Electrocution can kill,” “Can cause burns,” “Cuts can occur,” or “Can lead to data loss,” for example, will tell your readers the results stemming from the dangers, warnings, or cautions.

Avoidance steps. In three to five words, tell the readers how to avoid the consequences noted: “Wear rubber shoes,” “Don’t touch until cool,” “Wear protective gloves,” or “Keep disks away.”

Components of Instructions

- **Icons.** Your hazard alert should contain an icon—a picture of the potential consequence—to help everyone understand the caution, warning, or danger.
- **Table of Contents.** Your instruction might have several sections.
- In addition to the actual steps, the instructional manual could include technical specifications, warranties, guarantees, FAQs, troubleshooting tips, and customer service contact numbers.
- An effective table of contents will allow your readers to access any of these sections individually on an as-needed basis.

FIGURE 3 Hazard Alert

1	Hazard alert word
2	Hazard alert color
3	Alert symbol
4	Identification—what is the specific hazard?
5	Consequence—what are the potential results?
6	Avoidance steps—what should you do?
7	Hazard icon

Components of Instructions

- **Introduction.** Therefore, instructions often are reader-friendly and seek to achieve audience recognition and audience involvement. The manuals try to reach customers in a personalized way.

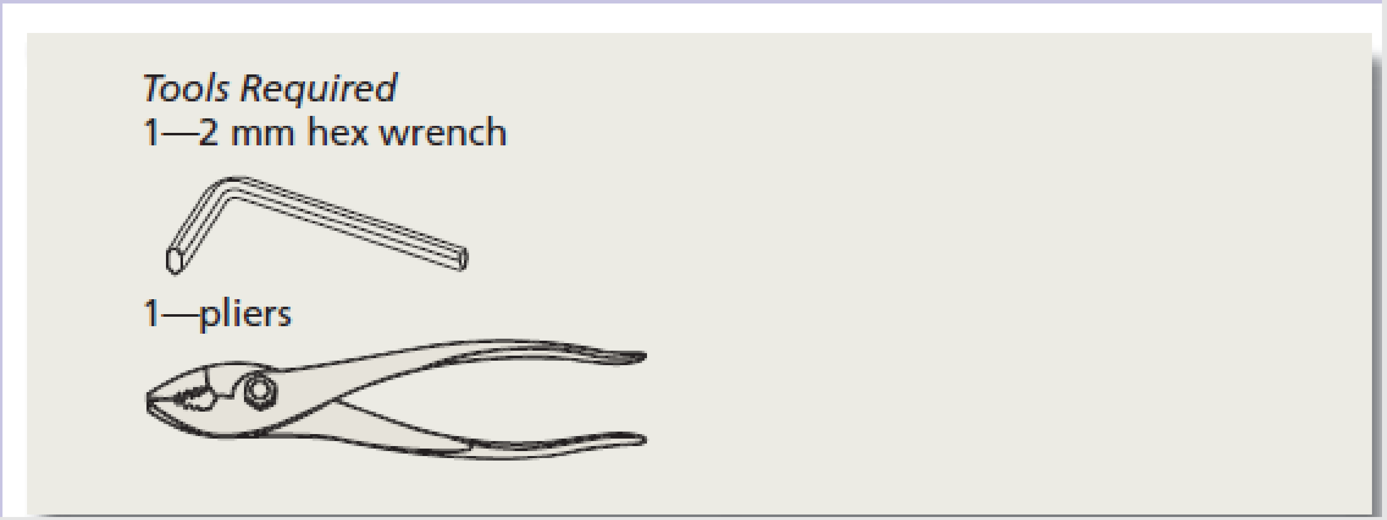
GLOSSARY

BDC	Bottom dead center
CCW	Counterclockwise
Danger	This hazard alert designates the possibility of death. Be extremely careful when performing an operation.
RMS	Root mean square

- **Glossary.**

Thank you for your purchase. Installation and operating procedures are contained in this manual about your new 3-D TV. The minutes you spend reading these instructions will contribute to hours of viewing pleasure.

Components of Instructions



- **Required Tools or Equipment.**

Components of Instructions

BEFORE	AFTER
After rotating the discs correctly, grease each with an approved lubricant.	1. Rotate the disks clockwise so that the tabs on the outside edges align. 2. Lubricate the discs with 2 oz of XYZ grease.

BEFORE	AFTER
Overloaded Steps	Separated Steps
Start the engine and run it to idling speed while opening the radiator cap and inserting the measuring gauge until the red ball within the glass tube floats either to the acceptable green range or to the dangerous red line.	1. Start the engine. 2. Run the engine to idling speed. 3. Open the radiator cap. 4. Insert the measuring gauge. 5. Determine whether the red ball within the glass tube floats to the acceptable green range or up to the dangerous red line.

Verbs Begin Steps

1. *Organize* the steps chronologically.
2. *Number* the steps.
3. *Use* highlighting techniques.
4. *Limit* the information within each step.
5. *Develop* the points thoroughly.
6. *Use* short words and phrases.
7. *Begin* the steps with verbs

BEFORE	AFTER
1. Press right arrow button to scroll through list of programs. 2. Select program to scan. 3. Place item to scan face down on scanner glass in upper left corner.	1. Press the right arrow button to scroll through a list of programs. 2. Select the program you want to scan. 3. Place an item to scan face down on the scanner glass in the upper left corner.

- **Instructional Steps.**
  - **Organize the Steps Chronologically.**
  - **Number Your Steps.**
  - **Use Highlighting Techniques.**
  - **Limit the Information within Each Step.**
  - **Develop Your Points Thoroughly.**
  - **Use Short Words, Sentences, and Paragraphs.**
  - **Begin Your Steps with Verbs.**
  - **Do Not Omit Articles.**

Additional Components

Smartphone Specifications

Height:	114 mm (4.48 inches)
Width:	66 mm (2.6 inches)
Depth:	15 mm (0.59 inches)
Weight:	136 grams (4.8 ounces)
Monitor Resolution:	480 × 320 pixel color display

- **Technical descriptions.**
- **Warranties.**
- Disclaimers - another common part of warranties that protect the manufacturer.

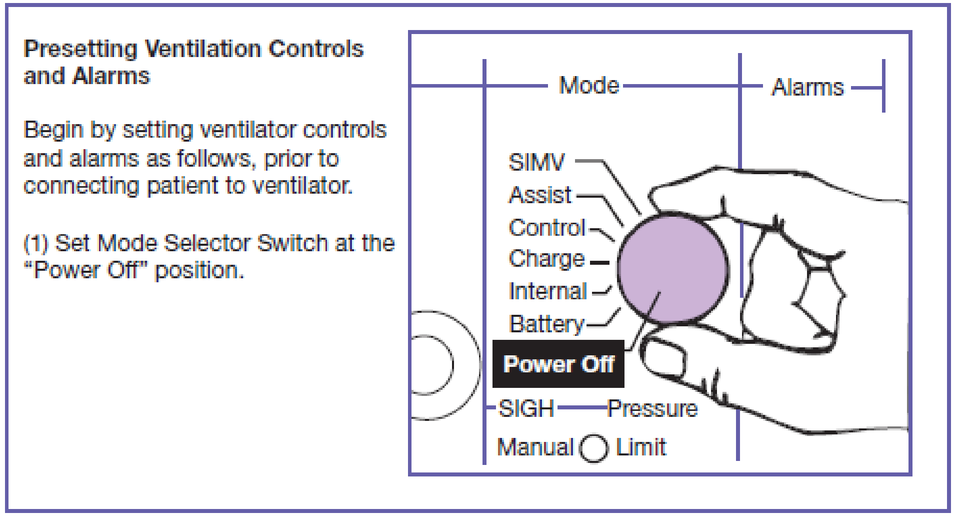
➢ **For example:**

- Note: Any changes or modifications to this system not expressly approved in this manual could void your warranty.
- Proof of purchase with a receipt clearly noting that this unit is under warranty- must be presented.
- The warranty is only valid if the serial number appears on the product.
- The manufacturer of this product will not be liable for damages caused by failure to comply with the installation instructions enclosed within this manual.

TABLE 1 Text without Graphics for High-Tech Readers

Location Item	Action	Remarks
Note: For additional operating instructions, refer to Companion 2800 operator's manual.		
1. MODE switch	Set to POWER OFF.	
2. Rear panel	Ensure that instructions on all labels are observed.	
3. Patient tubing circuit	Assemble and connect to the unit.	See Table 2, step 3.
4. Power cord	Connect to a 120/220 V AC grounded outlet.	If the power source is an external battery, connect the external battery cable to the unit per Table 2, step 4.
5. Cascade I humidifier	Connect to the unit.	See Table 2, step 2.

FIGURE 4 Instructions with Large, Bold Graphics for Low-Tech Audience



Additional Components

- **Accessories. (**an accessories list offers customers equipment such as extra-long cables or cords, carrying cases, long-life rechargeable batteries, belt clips and cases, wireless headsets, automotive chargers, and memory cards. Often the specifications are also provided for these accessories.)
- **Frequently asked questions.**
- **Corporate contact information.**
- **Graphics.**

Collaboration to Create User Manuals

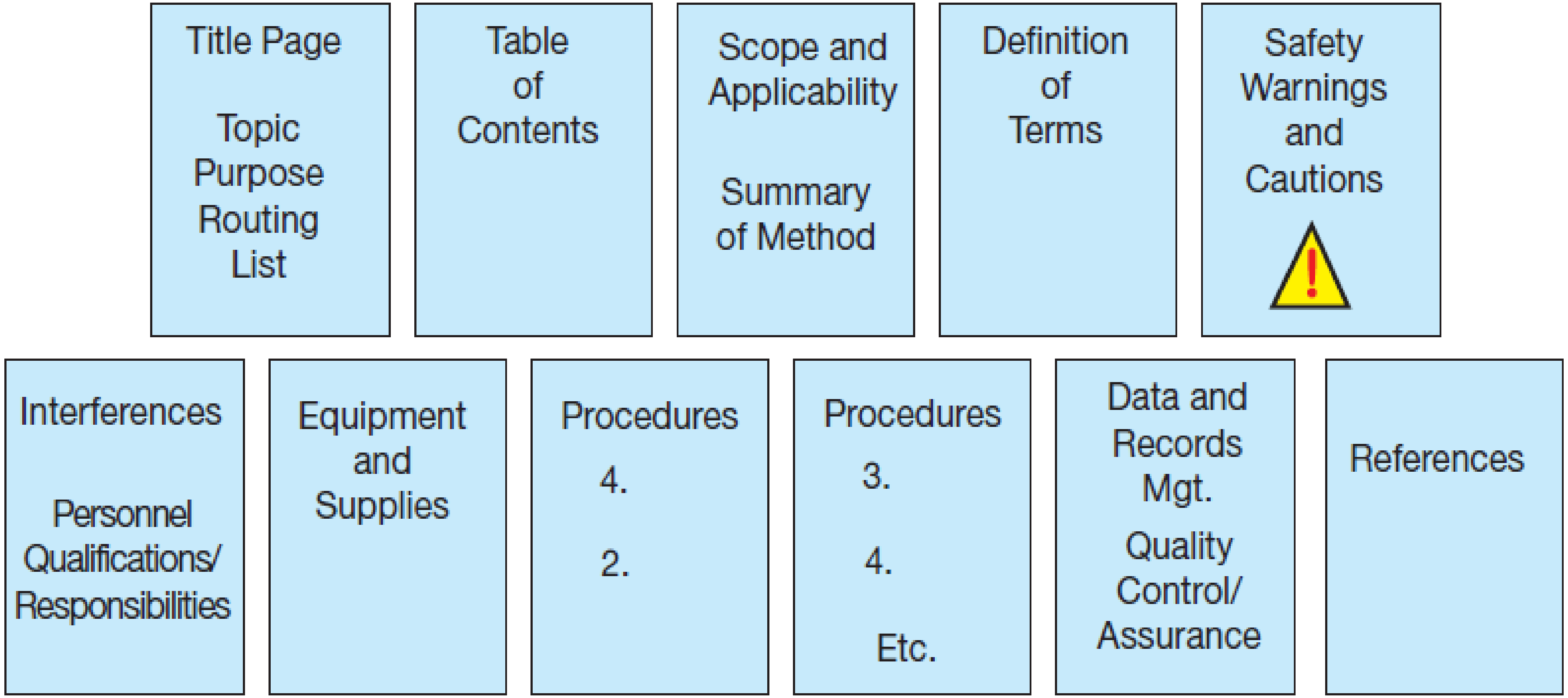
Instructional Videos

Standard Operating Procedures

- An SOP is a set of written instructions that documents routine or repetitive technical or administrative activities followed by business and industry.
- **Why Write an SOP?**
- Calibrate and standardize instruments
- Collect lab samples
- Handle and preserve food
- Analyze test data
- Troubleshoot equipment, machinery, and procedures
- List mathematical steps to follow for acquiring data and making calculations
- Assess hardware and software analytical data

Components of SOPs

FIGURE 6 Key Components of SOPs



Components of SOPs

- Title page
- Scope and Applicability.

Standard Operating Procedure

January 14, 2014
Preparing and Processing Algae Samples to Ensure Proper pH

Author: ______________________
Date Approved: ________________
Manager: _____________________
Date Approved: ________________
Quality Assurance Manager: _____
Date Approved: ________________

Scope

The purpose of this SOP is to establish uniform procedures for water compliance inspections (WCIs) performed by the Georgia Science and Assessment Division (GSAD). The WCIs evaluate the effectiveness and reliability of the state agency's inspection procedures to meet regulations of the Clean Water Act (CWA). The SOP will ensure thoroughness of all water compliance inspections and reports. The inspector may alter SOPs due to unexpected or unique problems in the field. Deviations from the SOP must be reported.

Applicability

The policies and procedures of the SOP apply to all personnel who take part in the WCIs.

Components of SOPs

- **Summary of Method.**
  - Title of people involved
  - Their roles and duties
  - Sequence of their involvement
  - Sequence of activities performed
- **Interferences.**
  - Humidity
  - Temperature
  - Depth
  - Altitude
  - Weather
  - Cleanliness
  - Carelessness
  - Sample size
  - Contamination

> *Interferences*
>
> If the water has high concentrations of chlorine, this can interfere with test results. Therefore, if the inspector determines that chlorine is present, add sodium sulfide to the sample bottle before autoclaving.

Components of SOPs

- **Personnel Qualifications/Responsibilities.**
- **Equipment and Supplies.**
- **Data and Records Management.**
  - Calculations to be performed during the procedure
  - Forms for the reports
  - Required reports
  - Reporting intervals
  - Report recipients
  - Process to follow for recording and storing data and information generated by the SOP
- **Quality Control and Quality Assurance.**
- **References.**

Test for Usability

- **Select a test audience.**
- **Ask the audience to test the instructions.**
- **Monitor the audience.**
- **Time the team members.**
- **Quantify the audience’s responses.**

INSTRUCTION, USER MANUAL, AND SOP USABILITY CHECKLIST

Audience Recognition

_____ **1.** Are technical terms defined?

_____ **2.** Are graphics used to explain difficult steps at the reader's level of understanding?

_____ **3.** Do the graphics depict correct completion of difficult steps at the reader's level of understanding?

_____ **4.** Are the tone and word usage appropriate for the intended audience?

_____ **5.** Does the introduction involve the audience, clarify how the reader will benefit, or provide the scope and applicability of the procedure?

Development

_____ **1.** Are steps precisely developed?

_____ **2.** Is all required information provided, including hazards, technical descriptions, warranties, accessories, and required equipment or tools?

Conciseness

_____ **1.** Are words, sentences, and paragraphs concise and to the point?

_____ **2.** Does each step specify one action versus overloading the reader?

Consistency

_____ **1.** Is a consistent and parallel hierarchy of headings used?

_____ **2.** Are graphics presented consistently (same location, same use of figure titles and numbers, similar sizes, etc.)?

_____ **3.** Does wording mean the same throughout (technical terms, cautions, warnings, notes, etc.)?

_____ **4.** Is the same system of numbering used throughout?

Ease of Use

_____ **1.** Can readers easily find what they want through the following?

- Table of contents
- Glossary
- Hierarchical headings
- Headers and footers
- Index
- Cross-referencing or hypertext links
- Frequently asked questions (FAQs)

Document Design

_____ **1.** Do graphics depict how to perform steps?

_____ **2.** Is white space used to make information accessible?

_____ **3.** Does color emphasize hazards, key terms, or important parts of a step?

_____ **4.** Do numbers or bullets divide steps into manageable chunks?

_____ **5.** Are hazards (dangers, warnings, cautions, and notes) boldfaced, typed in appropriate colors, and/or placed in textboxes to emphasize importance?

Activity

- Using any machine or software program you know well, create a flow chart for the sequences you want a reader to learn. Convert that flow chart into step-by-step instructions.